

BusBridge AXI4-Lite IP

Minor Project Report

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AXI4-Lite • SystemVerilog • Icarus Verilog • GTKWave

Verification Result

8 Tests Passed • 0 Tests Failed

Public GitHub Repository

<https://github.com/kolejayesh18/BusBridge-AXI4-Lite-IP>

1. Project Overview

BusBridge is an AXI4-Lite based peripheral IP block providing a memory-mapped register interface for control, status, data, and interrupt functionality. The project covers RTL implementation, protocol-oriented verification, simulation, waveform inspection, and integration documentation.

2. Problem Statement

- Develop a reusable AXI4-Lite peripheral IP block with clear register access behavior.
- Support independent AXI write address and write data channels.
- Provide deterministic read/write responses, including SLVERR for invalid addresses.
- Add event-driven interrupt and status handling.

3. Objectives

- Implement AXI4-Lite read and write channels in SystemVerilog.
- Create control/status register logic.
- Support byte write strobes and memory-mapped accesses.
- Verify normal transactions, error responses, interrupt behavior, and AW/W independence.
- Package the design with documentation and waveform evidence.

4. Project Requirements and Phases

- Phase 1 — Register and Protocol Planning: define register map and AXI-Lite read/write behavior.
- Phase 2 — RTL Integration: create peripheral modules and interrupt/status signaling.
- Phase 3 — Protocol Verification: build a verification testbench and validate corner cases/error responses.
- Phase 4 — Packaging and Reuse: provide integration documentation, constraints, and verification evidence.

5. System Architecture

- AXI4-Lite master connects to the BusBridge AXI4-Lite slave interface.
- Write address and write data are captured independently before a register write is issued.
- Read addresses are accepted and corresponding register data is returned through the AXI read channel.
- The register block produces ready_o and irq outputs and consumes event_i.

6. AXI4-Lite Interface

- Write Address: AWADDR, AWVALID, AWREADY.
- Write Data: WDATA,WSTRB, WVALID, WREADY.
- Write Response: BRESP, BVALID, BREADY.
- Read Address: ARADDR, ARVALID, ARREADY.
- Read Data: RDATA, RRESP, RVALID, RREADY.
- Separate AW and W storage allows the channels to arrive independently.

7. Register Map

Address	Register	Description
0x00	CTRL	Control register
0x04	STATUS	Status register
0x08	DATA	Data register
0x0C	IRQ_EN	Interrupt enable
0x10	IRQ_STATUS	Interrupt status

Valid accesses return OKAY; unsupported addresses return SLVERR.

8. RTL Implementation

- rtl/busbridge_axi_lite.sv — AXI4-Lite interface and transaction control.
- rtl/busbridge_regs.sv — memory-mapped register block and peripheral behavior.
- Write address/data are buffered using pending flags.
- AXI responses are generated after complete transactions.
- Reset is active-low through rst_n.

9. Verification Methodology

- tb/tb_busbridge.sv drives AXI4-Lite transactions and checks responses and data.
- Tests cover register access, interrupt behavior, invalid-address responses, and independent AW/W channels.
- Simulation was performed with Icarus Verilog.
- A VCD waveform was generated and inspected using GTKWave.

10. Simulation Results

- Simulation executable: sim/busbridge.vvp
- Waveform database: sim/busbridge.vcd
- Waveform evidence: reports/waveform.png
- Simulation completed with 8 passed tests and 0 failures.

11. Verification Results

Test	Result
CTRL write/read	PASS
DATA write/read	PASS
IRQ assertion after event	PASS
IRQ status set	PASS
IRQ clear	PASS
Invalid write returns SLVERR	PASS
Invalid read returns SLVERR	PASS
Independent AW/W channels	PASS
TOTAL	8 PASS / 0 FAIL

12. Waveform Evidence

The generated VCD waveform was opened and inspected in GTKWave. The waveform screenshot is included in the repository as reports/waveform.png.

Evidence file: reports/waveform.png

13. Repository Structure

- README.md — project overview and author information.
- project_requirements/ — original project details and requirements.
- rtl/ — SystemVerilog RTL source.
- tb/ — verification testbench.
- sim/ — simulation artifacts.
- reports/ — verification report and waveform evidence.
- docs/ — integration guide.
- constraints/ — constraints.
- scripts/ — project scripts.
- presentation/ — project PowerPoint presentation.

14. Conclusion

The BusBridge AXI4-Lite IP project was implemented and functionally verified. The final simulation completed with 8 tests passed and 0 tests failed. Verification covered register access, interrupt behavior, invalid-address error responses, and independent AXI AW/W channels. The repository contains the requirements, source code, testbench, simulation evidence, verification documentation, integration guidance, and presentation.

15. GitHub / Live Project

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Public Repository: <https://github.com/kolejayesh18/BusBridge-AXI4-Lite-IP>

Project scope note: Vivado synthesis/implementation and lint were not performed in this project run and are therefore not claimed in this report.